

Excercise 3 and 4

## Mocking and Stubbing

MyServiceTest.java

```
MyServiceTest.java x  AssertionsTest.java  CalculatorAAATest.java
2
3  import org.junit.Test;
4
5  import static org.junit.Assert.assertEquals;
6  import static org.mockito.Mockito.*;
7
8  public class MyServiceTest {
9
10
11     @Test
12     public void testExternalApi() {
13
14         // Step 1 : Create Mock
15         ExternalApi mockApi = mock(ExternalApi.class);
16
17         // Step 2 : Stub
18         when(mockApi.getData()).thenReturn("Mock Data");
19
20         // Step 3 : Inject Mock
21         MyService service = new MyService(mockApi);
22
23         // Step 4 : Call Method
24         String result = service.fetchData();
25
26         // Step 5 : Verify Result
27         assertEquals("Mock Data", result);
28     }
29
```

```
Run  MyServiceTest x
[Icons] [Icons] [Icons] [Icons] [Icons] [Icons] [Icons] [Icons] [Icons] [Icons] [Icons] [Icons]
v  ✓ MyServiceTest (org.example) 873 ms  ✓ 1 test passed
    ✓ testExternalApi 873 ms
    "C:\Program F
    WARNING: A Ja
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    Java HotSpot
    Process finis
```

Verification and interaction of the test

```
@Test
public void testVerifyInteraction() {

    ExternalApi mockApi = mock(ExternalApi.class);

    when(mockApi.getData()).thenReturn("Mock Data");

    MyService service = new MyService(mockApi);

    service.fetchData();

    verify(mockApi).getData();
}
```

